ITAI 1378

Computer Vision for Artificial Intelligence

Module 07

Generative Models for Vision

The user perspective: how CV engineers use pretrained text to image and image to image models as workforce tools.

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Welcome to Module 07

The first six modules of this course taught you how to analyze images. You learned to classify them, detect objects in them, and segment them at the pixel level. Module 07 turns the arrow around. This module is about generating images, not analyzing them. You will learn how to use modern generative models like Stable Diffusion, FLUX, and DALL-E 3 as practical tools in CV pipelines.

This module is deliberately user focused, not builder focused. The mathematics of diffusion models, the architecture of the noise prediction network, and the techniques for training a model from scratch all live in ITAI 2376 (Deep Learning), where they belong with the broader generative AI curriculum. Here in ITAI 1378 we focus on what every working CV engineer actually does day to day: load a pretrained model, give it instructions, get useful images back, and integrate the result into a larger system.

You will leave this module able to do four things. First, you will know the model landscape (open versus closed, the major families, when to use which) well enough to pick the right tool for any job. Second, you will know how to write prompts and tune sampling parameters to get the images you want. Third, you will master the image to image workflows that matter most in production: img2img, inpainting, ControlNet, and super resolution. Fourth, you will understand the limitations, ethical concerns, and legal landscape well enough to use these tools responsibly.

Like every booklet in this course, this is an evergreen reference. The specific model names will evolve. Stable Diffusion 4 may exist by the time you read this. FLUX may have a successor. The concepts (what prompts do, what guidance scale means, why synthetic data is risky in training) will remain the same. Treat this booklet as a desk reference for the working CV engineer, not a study guide for a single test.

KEY TAKEAWAY | What this module will give you

Practical fluency with the generative AI tools every CV engineer is expected to use in 2026. Hugging Face Diffusers, text to image, img2img, inpainting, ControlNet, super resolution, and the synthetic data workflows that solve the most common problem in production CV: not enough labeled data. Plus the judgment to know what these tools cannot do and the awareness to use them ethically.

Part 01 | What Generative Models Do for CV Engineers

Before diving into specific tools, this part frames what generative models actually deliver for CV engineers. We discuss what makes them different from the analysis models in earlier modules, the four workforce use cases that matter most, and a brief history that puts current models in context.

Analysis Versus Generation

Every model in Modules 01 through 06 took an image as input and produced some kind of information as output. Classification returned a label. Detection returned bounding boxes. Segmentation returned masks. Generative models reverse this flow. They take text or another image as input and produce a new image as output. The CV pipeline runs in the opposite direction.

This reversal is conceptually simple but practically transformative. A detector tells you whether a defective part is present on a conveyor belt. A generative model can produce hundreds of synthetic examples of defective parts to train that detector when you only have a few real examples. The analysis tool and the generative tool work together, with the generative tool feeding the analysis tool's training data.

> ⚠ **MISCONCEPTION ALERT | Generative models do not replace analysis models**
> It is easy to fall into thinking generative AI replaces traditional CV. It does not. The CV problems your customers care about (counting parts on a line, finding tumors in scans, recognizing faces at a security checkpoint) are still analysis problems and still need analysis models. Generative models are powerful complementary tools, not replacements. The skill is knowing when to reach for each.

The Four Workforce Use Cases

Across industry CV teams in 2026, four use cases of generative models dominate. Understanding these is how you connect generative AI to real business value.

Use Case 1: Synthetic Training Data

Labeled training data is the most expensive bottleneck in CV. Generative models can dramatically expand a small labeled dataset by producing variations of existing examples. If you have 100 photos of a manufacturing defect, you can generate 1000 more in different lighting, angles, and presentations. The resulting trained classifier becomes more robust to real world variation.

Use Case 2: Image Editing and Restoration

Inpainting damaged photos. Removing backgrounds for product catalogs. Super resolution for low quality cameras. Cleaning up privacy sensitive elements from datasets. These tasks once required custom CNN training or expensive manual work. With pretrained generative models, they become quick function calls.

Use Case 3: Rapid Prototyping

Need a hero image for a product mockup? Marketing imagery for a campaign? A reference rendering for a robotics simulation? Generative models produce decent first drafts in seconds. The output is rarely the final asset, but it accelerates the early prototype phase enormously.

Use Case 4: Data Augmentation

Closely related to synthetic training data but more subtle. Generative augmentation creates new training examples that preserve labels but vary lighting, weather, time of day, or artistic style. Models trained with this kind of augmentation handle domain shifts in production much better than models trained on a single condition.

INDUSTRY PRACTICE | Where the money is

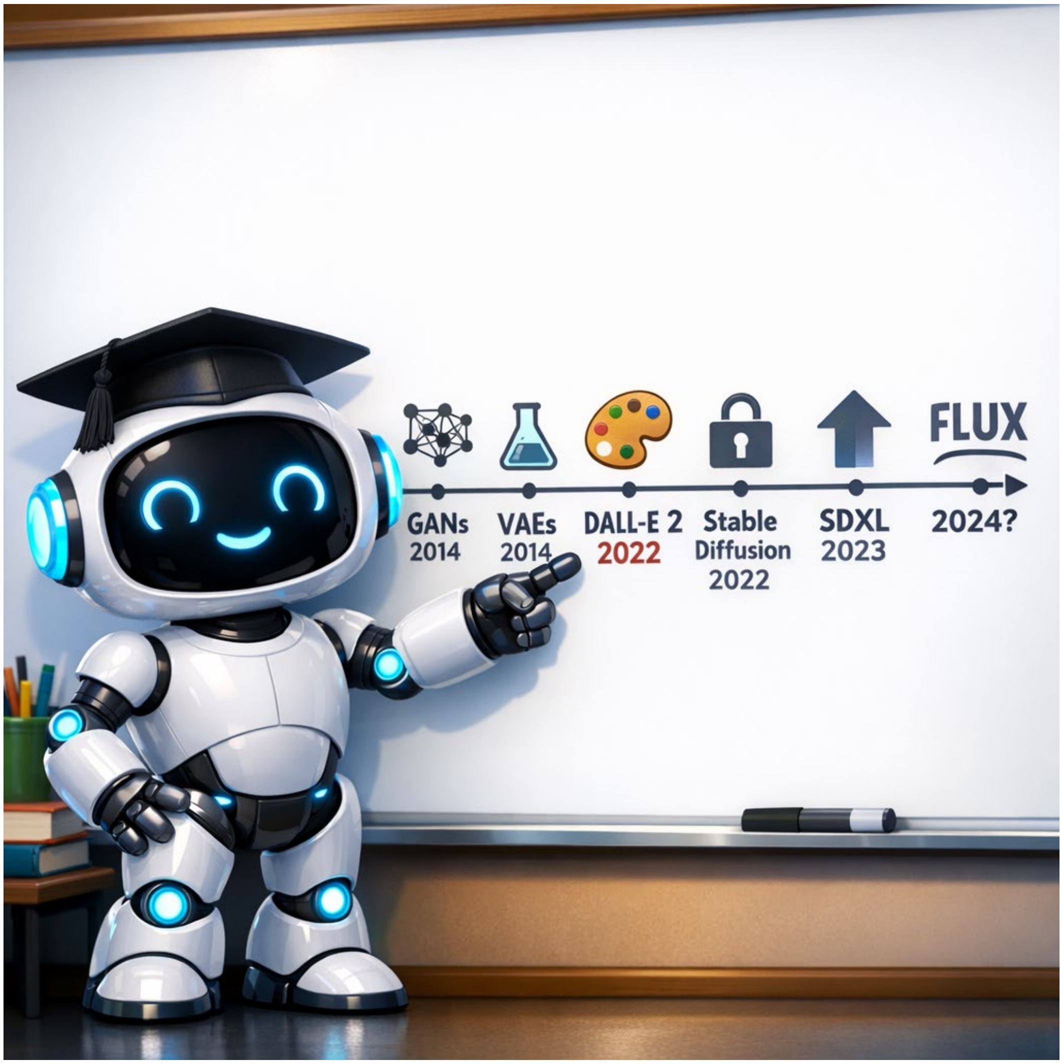
Of these four use cases, synthetic training data and augmentation are the highest impact for industry CV teams. They directly solve the bottleneck (labeled data) that limits most projects. Editing and prototyping are visible and fun but lower revenue impact. When you talk to hiring managers about generative AI experience in a CV context, lead with synthetic data work.

A Brief History

Generative models for images have a roughly twelve year arc. Generative Adversarial Networks (GANs), introduced by Ian Goodfellow in 2014, produced the first convincing fake faces and dominated research through the late 2010s. They were powerful but hard to train, and the output had limited diversity.

In 2022, two events reshaped the field. OpenAI released DALL-E 2, a text to image model that brought generative AI to public attention. The same year, Stability AI released Stable Diffusion as open weights, making generative models freely runnable on consumer hardware. The combination of accessibility and capability triggered a wave of innovation.

Since then, the pace has accelerated. Stable Diffusion XL (SDXL) raised the quality bar in 2023. Black Forest Labs released FLUX in 2024, fixing the long standing 'AI cannot draw hands or text' problems. Closed models like DALL-E 3, Midjourney, and Google Imagen continue to evolve in parallel. By 2026, generative image models are no longer a research novelty; they are a routine workforce tool.



KEY TAKEAWAY

Generative models for vision reverse the CV pipeline: text or image in, new image out. Four workforce use cases dominate: synthetic training data, image editing and restoration, rapid prototyping, and data augmentation. The field grew from GANs in 2014 to production tools by 2024. In 2026, this is workforce baseline.

KNOWLEDGE CHECK | Quick check on framing

1. How do generative models differ from the analysis models in Modules 01 through 06?
2. Name the four workforce use cases and give one industry example of each.
3. Why is synthetic training data considered the highest impact use case for industry CV teams?

Part 02 | The Model Landscape

This part covers the landscape of generative image models available to CV engineers in 2026. You will learn the big distinction (open weights versus closed APIs), the major families in each camp, the practical tradeoffs that decide which to use, and the Hugging Face Diffusers library that has become the standard interface for open models.

Open Weights Versus Closed APIs

Generative image models come in two major flavors based on how you access them. Open weights models are downloadable; you get the trained parameters and run them on your own hardware. Closed API models are services; you call an endpoint with your prompt and receive an image back over the network.

Open models give you control. Run them locally, no API costs, full reproducibility, modify them freely. Closed models give you convenience. No GPU required, no setup, often the highest quality. The choice depends on your scale, your privacy needs, your budget, and your customization requirements.

DEEPER LOOK | When open weights pay off in dollars

A rough rule. Closed API services charge per image, typically a few cents to a few dimes each. Open models run on a GPU that you rent or own. The breakeven depends on usage. Generating one image a day: closed APIs are cheaper. Generating thousands per hour for production: open weights win, often by an order of magnitude. The numbers shift constantly; do the math for your specific workload.

Stable Diffusion: The Open Workhorse

Stable Diffusion, developed by Stability AI and released as open weights in 2022, is the foundational open model in the field. Several major versions exist.

Stable Diffusion 1.5

The breakout version from 2022. Small enough to run on a consumer GPU with 8GB of VRAM. Tens of thousands of fine tuned variants exist for specialized aesthetics or domains. Still useful for projects where speed matters more than quality.

Stable Diffusion XL (SDXL)

Released in 2023. Bigger model, higher quality output, native support for 1024 by 1024 resolution. Became the workforce default for open generation in 2024. Requires more VRAM (around 12GB minimum) but the quality jump is substantial.

Stable Diffusion 3 (SD3)

The current official version. Larger and more capable than SDXL. Improved understanding of compositional prompts. Better text rendering. The trade-off is higher compute requirements.

FLUX: The Modern Open Alternative

FLUX, released by Black Forest Labs in 2024, is the most recent serious challenger to Stable Diffusion in the open model space. The team behind it includes several original Stable Diffusion authors. FLUX was specifically designed to fix the well known weaknesses of earlier diffusion models: garbled text in generated images, distorted hands, and poor understanding of compositional prompts.

Two FLUX variants are most common. FLUX.1 [dev] is the open weights version intended for serious work; it produces the highest quality but requires a powerful GPU. FLUX.1 [schnell] is a smaller distilled version that runs faster, is free for commercial use, and is the right default for most workforce experimentation.

INDUSTRY PRACTICE | What to install on day one

If you are setting up a CV development workstation in 2026, install the Hugging Face Diffusers library, download FLUX.1 [schnell] and SDXL, and try each on the same prompts. Get a sense of how they differ in style and capability. These two together cover the vast majority of workforce open model use cases. Add a ControlNet checkpoint for spatial control. That kit handles 80 percent of CV engineering work involving generation.

Closed Models: API Services

Three major closed model providers dominate the API space.

DALL-E 3 (OpenAI)

Released in late 2023. Strong prompt understanding because it uses a large language model to interpret the user's prompt before generating. Distinctive ability to handle complex compositional requests. Available through OpenAI's API and integrated into ChatGPT. Pricing per image at the time of writing is comparable to other premium services.

Midjourney

A long running closed service with a distinctive aesthetic. Strong in artistic and design oriented generation. Historically available only through Discord; web access opened in more recent versions. Popular among designers and creatives. No traditional API for most of its history, though that has slowly evolved.

Imagen (Google) and Firefly (Adobe)

Imagen powers Google's image generation features across Search, Workspace, and Vertex AI. Adobe Firefly is the major commercial alternative, with the distinguishing feature that Adobe trains it exclusively on licensed and public domain content. Firefly carries an indemnification guarantee for commercial use, making it the safest choice when copyright clarity matters.

CAREER CONNECTION | Which APIs to mention in interviews

In a CV engineer interview in 2026, knowing the names DALL-E 3, Midjourney, Imagen, and Firefly is workforce baseline. Knowing the open alternatives (Stable Diffusion variants and FLUX) is also baseline. Knowing the tradeoffs (open versus closed, when each is appropriate) is the difference between baseline and strong. Practice articulating: 'For high volume production at controlled cost, I would use open models with Diffusers. For client deliverables with copyright safety, I would use Firefly. For complex prompts where understanding matters most, DALL-E 3.'

The Hugging Face Diffusers Library

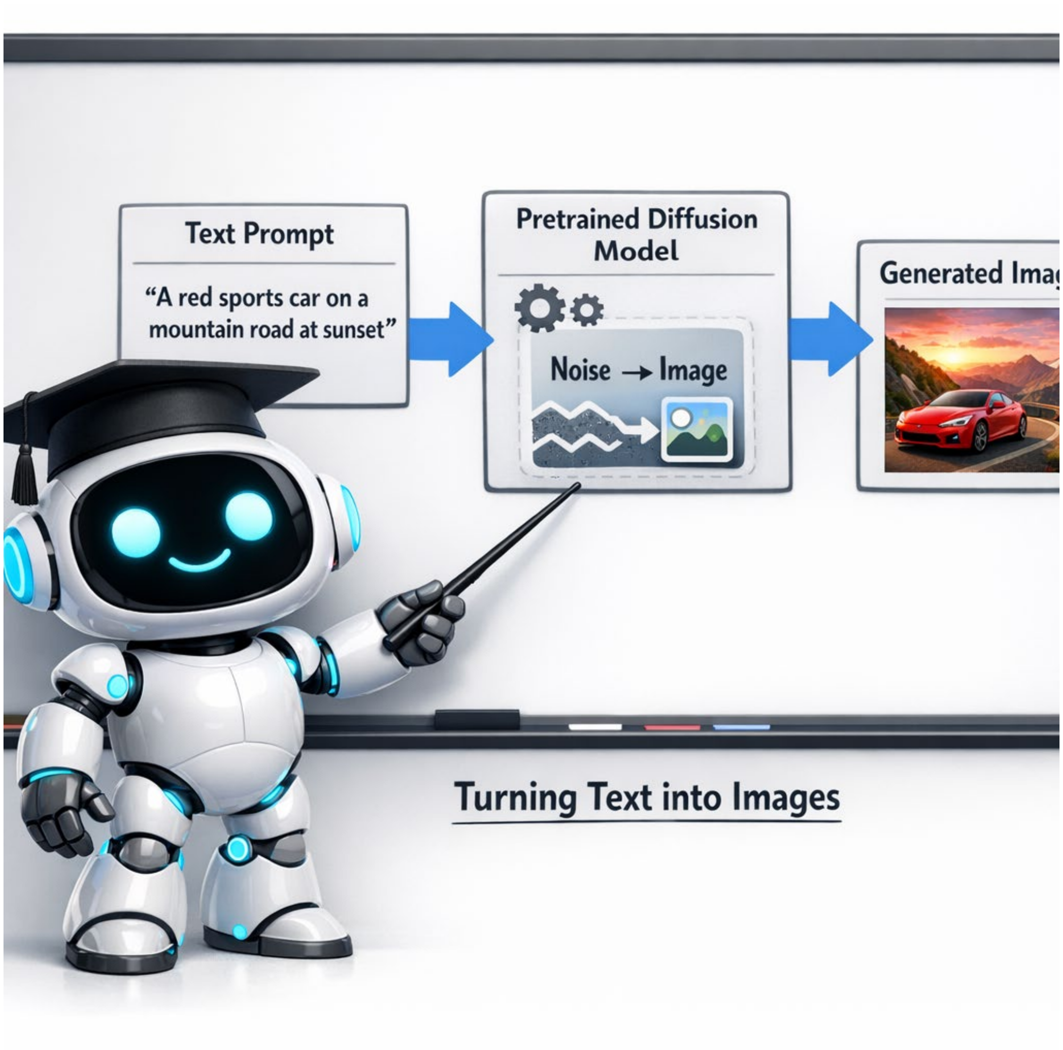
If you remember one library name from this module, make it Diffusers. The Hugging Face Diffusers library has become the de facto Python interface for working with diffusion models. Open source, free, well maintained, and supports almost every open model worth using.

What Diffusers Gives You

A unified API across Stable Diffusion, SDXL, SD3, FLUX, and many smaller models. Pipelines for the major workflows: text to image, image to image, inpainting, ControlNet, image variation. Helper utilities for sampling schedulers, prompt encoding, and image saving. Integration with PyTorch and CUDA out of the box.

The Pipeline Abstraction

Diffusers organizes models as pipelines. A pipeline bundles together the model weights, the text encoder, the sampler, and the postprocessing logic into one Python object you can call. Loading a pipeline takes one line. Running it on a prompt takes another. The complexity is hidden but accessible if you need to dig in.



Choosing the Right Tool for the Job

Match the model to the use case. For client deliverables where copyright safety is paramount, Adobe Firefly is the leading choice. For high volume internal use where cost matters, run FLUX or SDXL via Diffusers on a local or rented GPU. For complex prompts where understanding subtle requirements matters more than raw quality, DALL-E 3. For artistic and design work, Midjourney has the distinctive aesthetic.

In practice, most CV engineering teams in 2026 settle on one open model (FLUX or SDXL) for high volume work plus one API service (typically DALL-E 3 or Firefly) for special cases. Knowing both flavors of the landscape lets you make informed tradeoffs.

KEY TAKEAWAY

The model landscape splits into open weights (Stable Diffusion family, FLUX) and closed APIs (DALL-E 3, Midjourney, Imagen, Firefly). Open models give you control, low cost, and reproducibility. Closed models give you convenience and sometimes higher quality. Hugging Face Diffusers is the standard library for open models; learn it. Match the model to the use case and budget.

KNOWLEDGE CHECK | Quick check on the model landscape

1. Name two open weights models and two closed API services, with one strength of each.
2. What is Hugging Face Diffusers, and what does the Pipeline abstraction give you?
3. A client needs commercially indemnified imagery with copyright safety. Which service would you reach for?

Part 03 | Text to Image and Prompt Engineering

This part covers the most basic generative workflow: text in, image out. You will learn how the call is structured, how to write prompts that produce the images you want, what the standard parameters control, and how to get reproducible results.

The Basic Call

At the simplest level, generating an image from text is three lines of Python with Diffusers. Load the pipeline, pass the prompt, save the result. The model handles the rest. This simplicity is what makes generative tools accessible: you do not need to understand the diffusion process to use it well, just like you do not need to understand internal combustion to drive a car.

In practice, real workflows wrap this basic call in a loop that varies parameters and saves multiple candidates. You generate, evaluate, adjust the prompt, generate again. The iteration cycle is fast (seconds to minutes per image) which makes prompt engineering a primarily experimental skill rather than an analytical one.

🔍 DEEPER LOOK | Where the magic happens (briefly)

Without diving into the math (which belongs in a different course), here is what happens behind that one line call. The pipeline encodes your text prompt into a numeric representation. It samples a starting point from random noise. It applies a sequence of denoising steps, each one nudging the noise closer to a coherent image that matches your prompt. The pretrained model knows how to do those denoising steps because of its training on millions of image text pairs. After enough steps, you have an image. The library handles every detail.

Prompt Engineering: Words That Work

A prompt is the program. The same model can produce wildly different outputs based on prompt phrasing alone. Five principles cover most of the craft.

Principle 1: Lead With the Subject

The most important content goes early. 'A golden retriever puppy lying on green grass, soft afternoon sunlight' beats 'Soft afternoon sunlight, green grass, a golden retriever puppy lying'. Models give more weight to early tokens.

Principle 2: Specify Style and Medium

Generic prompts produce generic images. Tell the model whether you want a photograph, a painting, a 3D render, a pencil sketch, a vintage poster. Name a specific style if you have one in mind: 'oil painting in the style of impressionism', 'professional product photography with studio lighting', 'colored pencil sketch on textured paper'.

Principle 3: Add Concrete Details

Specific is better than vague. 'A red sports car' is generic. 'A 1967 Ferrari 275 GTB in deep red, parked on a cobblestone street, rain reflections on the pavement, blue hour lighting' is concrete. Each added detail steers the output toward what you actually want.

Principle 4: Iterate One Variable at a Time

If your output is close but not quite right, change one thing about the prompt. Adjust the lighting, swap the medium, add or remove a detail. This makes it easy to learn which words drive which changes. Changing five things at once teaches you nothing.

Principle 5: Use Reference Examples

When you find a prompt structure that works for a given style, save it as a template. Most production prompt engineering involves a handful of templates that get adapted for specific scenes rather than starting from scratch each time.

INDUSTRY PRACTICE | How professionals organize their prompts

Production CV teams that use generative models build internal prompt libraries. Categorized by style, by subject, by intended use. They version control prompts the way developers version control code. When a prompt is found that produces consistent high quality output for a recurring need, it gets named and stored. This is one of the most underrated workflow improvements in generative AI work.

Negative Prompts

Some models, especially the Stable Diffusion family, accept a second prompt that describes what to avoid. Negative prompts are a powerful tool for steering away from common failure modes. Common entries: 'blurry, low resolution, distorted hands, extra fingers, watermark, text artifacts, oversaturated'. These hints help the model avoid recurring quality problems.

Not all models use negative prompts the same way. FLUX needs them much less than Stable Diffusion because its training data was cleaner. DALL-E 3 does not use negative prompts at all; its language model handles avoidance internally. Vocabulary point: students should know what a negative prompt is even on models where it is not relevant.

THINK ABOUT IT | When the negative prompt is the real prompt

Sometimes the easiest way to express what you want is to say what you do not want. If your generated portraits keep producing oversaturated faces, adding 'oversaturated colors' to the negative prompt may help more than rewriting the positive prompt. If your product photos keep generating odd shadows, 'harsh shadows, vignetting' in negative helps. Treat the negative prompt as a regular tool, not a last resort.

The Three Knobs: Steps, Guidance, Seeds

Three parameters control most of the practical behavior of a generative model. Master these and you have 95 percent of the practical control most workflows need.

Sampling Steps

How many refinement passes the model takes between starting noise and final image. More steps means higher quality and longer generation time. Typical defaults are 20 to 50 steps. For prototyping where speed matters, drop to 10 to 15. For final delivery where quality matters, raise to 50 to 80. The quality curve is steep at first and then flattens; you rarely benefit from more than 100 steps.

Guidance Scale

How strictly the model follows your prompt versus how naturally it generates. Higher guidance means the output sticks closer to the literal prompt. Lower guidance means the output is more natural but may stray from your intent. Typical values are 5 to 9 for SD family, with FLUX preferring lower values around 3 to 5. Above 10 or 15, images become oversaturated and unnatural. Below 2, output becomes loosely related to the prompt.

Seeds

The random starting point for generation. Same prompt plus same seed equals the same image, every time. This is critical for reproducibility. When you find an output you like, save the seed alongside the prompt. When you iterate on a prompt, fix the seed so you can see the effect of prompt changes in isolation. Without a fixed seed, two prompt variations may differ for reasons other than the prompt.

> ⚠ **MISCONCEPTION ALERT | More steps does not mean better images**
>
> Beginners often assume that running with 200 sampling steps will produce dramatically better images than 50. It will not. The quality curve plateaus quickly. After about 50 steps for SDXL, additional steps are mostly wasted compute. Past 100, the diminishing returns are extreme. Save your time and budget; tune the prompt and seed instead.

Reproducibility in Practice

Anytime you produce generative output you might want to reproduce, store the trifecta: prompt, seed, and full set of generation parameters. Modern tools like ComfyUI and many Diffusers workflows save metadata directly inside the PNG file headers. This means anyone who has the image can extract the exact recipe that produced it.

For team work, reproducibility matters more than for individual exploration. When a teammate asks 'how did you produce this image,' a complete answer includes the prompt, the model checkpoint, the seed, the steps, the guidance scale, and any ControlNet or img2img settings. Document or save automatically.

> **KEY TAKEAWAY**
>
> Text to image generation is a function call: prompt in, image out. The prompt is the program; five principles (lead with subject, specify style and medium, add concrete details, iterate one variable at a time, use reference examples) cover most of the craft. Negative prompts steer away from failures (especially in Stable Diffusion). The three knobs (steps, guidance, seeds) control quality, prompt fidelity, and reproducibility. Save seeds with outputs that matter.

KNOWLEDGE CHECK | Quick check on text to image

1. Name the five prompt engineering principles described in this part.
2. Describe what each of the three main parameters (steps, guidance, seed) controls.
3. A teammate produces a great image but cannot reproduce it. What information do they need to save to make their workflow reproducible?

Part 04 | Image to Image Workflows

Text to image gets the attention, but image to image is where most workforce CV value comes from. This part covers the four major workflows where you start with an existing image and generate a transformed result: img2img, inpainting, ControlNet, and super resolution with restoration. Each one solves a different practical problem.

img2img: Transform an Existing Image

Starting from text alone, the model has total creative freedom. Starting from text plus an existing image (the img2img workflow), the model is anchored to that image. The output preserves something of the original while adapting toward what the prompt describes. This is the most common generative workflow in production because it lets you maintain consistency while making targeted changes.

The Strength Parameter

img2img has one additional knob beyond the text to image parameters: strength. Strength controls how much the model is allowed to deviate from the input image. Low strength (around 0.1 to 0.2) produces tiny tweaks barely visible to the eye. Medium strength (0.4 to 0.6) produces style changes while preserving subject identity. High strength (0.7 to 0.9) makes major changes and loses much of the original. Strength near 1.0 effectively ignores the input and acts like pure text to image.

Common img2img Use Cases

Restyle a real photo to look like a painting or sketch. Convert daytime scenes to nighttime or other times of day. Transform product photos to different aesthetics. Generate variations of a hero image while preserving its layout. Create training data variants that share the same labels.

Inpainting: Surgical Edits With a Mask

Inpainting is the precise scalpel of generative tools. You provide three inputs: an original image, a mask indicating which area to change, and a text prompt describing the desired result for that area. Everything outside the mask is preserved exactly. Only the masked region is regenerated.

How the Mask Connects to Module 06

Inpainting requires a mask. In Module 06 you learned multiple ways to produce masks: U-Net for semantic regions, Mask R-CNN for object instances, SAM for interactive segmentation. The natural pipeline: use a segmentation model to produce the mask, then pass that mask to an inpainting model along with a prompt. This composition is one of the most powerful patterns in modern CV. Detect or segment to localize, generate to modify.

High Impact Inpainting Workflows

Remove unwanted objects from photos (segment the object, inpaint with a prompt describing the background). Replace product labels in stock imagery for prototyping. Fix damaged areas in restored photos. Scrub privacy sensitive elements (faces, plates) from training datasets while preserving the rest of the image. Add new elements to existing scenes seamlessly.

CAREER CONNECTION | The detect plus inpaint pipeline is interview gold

If you can articulate the detect plus inpaint workflow clearly in an interview, you have signaled real CV engineering judgment. 'I would use SAM to produce a mask of the regions I want to modify, then pass that mask to Stable Diffusion inpaint with a prompt describing the desired replacement. This composition pattern is more flexible than training a custom model for the specific edit.' That kind of fluent answer is rare and stands out.

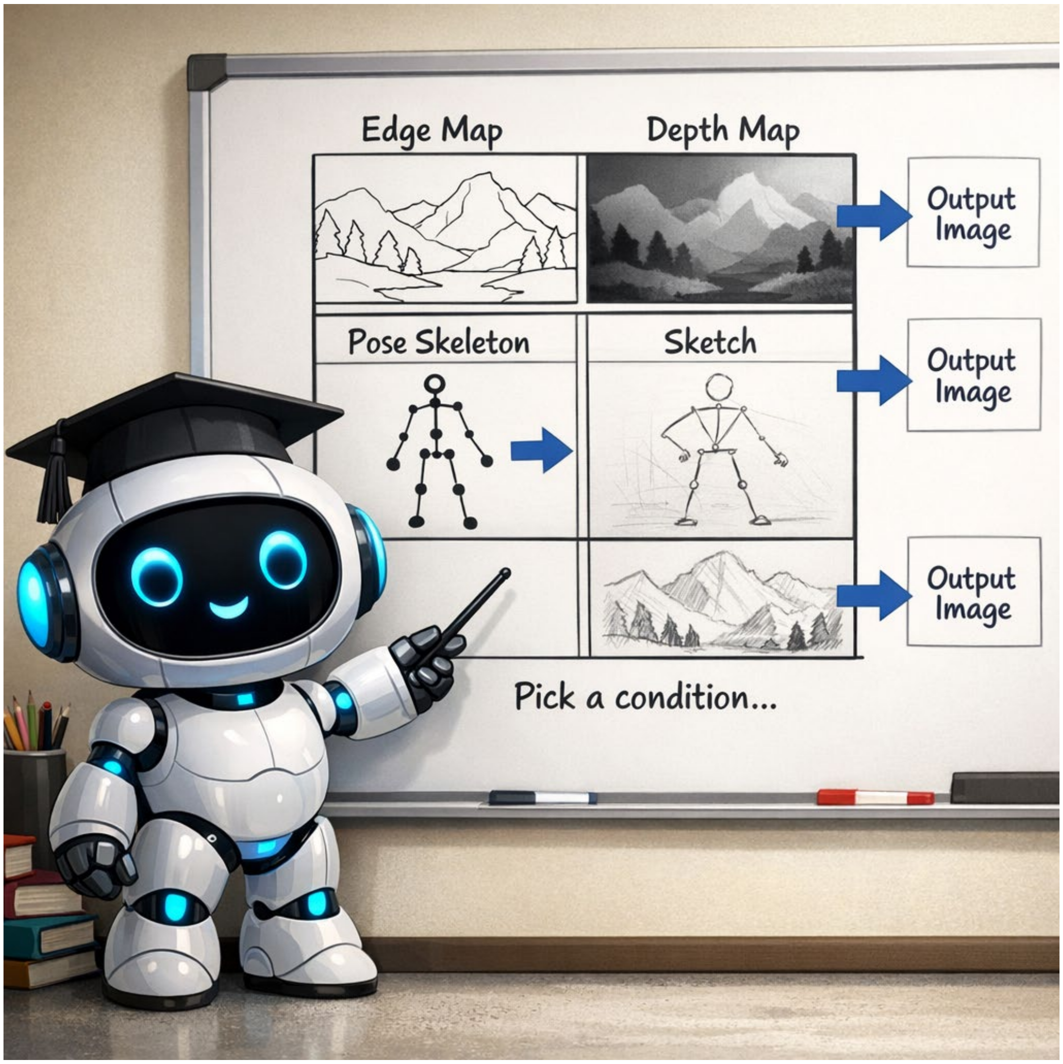
ControlNet: Guided Generation

Text alone cannot fully describe complex spatial layouts. ControlNet solves this by letting you condition generation on structural information from a guide image. You provide the model with both a text prompt and a structural map (an edge sketch, a depth map, a pose skeleton, or another conditioning signal). The output follows the structure of the guide while obeying the text.

The Common ControlNet Conditioning Types

Edge maps (Canny or HED). Trace the contours of a reference image and use them to guide layout. Good for preserving overall composition while changing style. Depth maps. Show the spatial layout of a scene in terms of distance. Useful for architectural and product visualization. Pose skeletons (OpenPose). Pin down the pose of human figures. Essential for character work and product photography with mannequins.

Sketches and scribbles. Rough doodles act as compositional guides. Good for converting whiteboard sketches into rendered concepts. Segmentation maps. Use a semantic segmentation map (you know how to produce these from Module 06) to specify what goes where. Useful for landscape and urban scene generation. Normal maps, line art, and several others exist as specialized cases.



Super Resolution and Restoration

Two specialized workflows that deserve mention because they once required custom CNN training and are now solved by pretrained models out of the box.

Super Resolution

Take a low resolution image and produce a higher resolution version with believable detail. Models like Real ESRGAN and SUPIR are the current workforce standards. Use cases include upscaling old photos for printing, sharpening surveillance footage for analysis (with awareness of forensic limitations), and improving images captured by low quality cameras.

Restoration

Repair old or damaged photos. Specialized models exist for different damage types. GFPGAN for face restoration. DeOldify for black and white colorization. Several denoising and deblurring models for motion or noise artifacts. These models combine techniques from generative AI and classical image processing in clever ways.

INDUSTRY PRACTICE | Restoration is sneaky high value

Restoration tasks rarely make headlines but generate enormous practical value. Museums and archives use them to digitally restore historical collections. Insurance companies use them to process scanned documents and damaged photos. Family genealogy services use them to bring old photos back to life. Knowing the names (Real ESRGAN, GFPGAN, SUPIR, DeOldify) and what they each do is a small but high impact piece of the workforce CV vocabulary.

KEY TAKEAWAY

Four image to image workflows dominate practical generative CV. img2img transforms existing images with a strength parameter controlling fidelity. Inpainting edits specific regions defined by a mask, composing naturally with segmentation tools from Module 06. ControlNet guides generation with structural information from a reference image. Super resolution and restoration use specialized pretrained models to upscale and repair. These are where most production value lives.

KNOWLEDGE CHECK | Quick check on image to image

1. What does the strength parameter control in img2img, and what is a typical value for style edits that keep subject identity?
2. How does inpainting compose with segmentation models from Module 06? Walk through the pipeline.
3. Name three ControlNet conditioning types and give a use case for each.

Part 05 | Synthetic Data and Reality Checks

This part is the heart of the workforce case for generative AI in CV. You will learn how to use generative models to expand small datasets, perform domain adaptation through style transfer, and avoid the pitfalls that trip up teams when they treat synthetic data as a replacement for real data rather than a complement to it.

The Labeled Data Problem

Almost every real world CV project hits the same wall eventually: not enough labeled data. Modern models need thousands of labeled examples per class for strong performance. Annotation is expensive (a few cents per classification label, dollars per bounding box, tens of dollars per segmentation mask). Many projects have dozens or hundreds of real examples when they need thousands.

Before generative AI, the standard responses to this problem were limited. Classical data augmentation (flipping, rotating, cropping, color shifts) helped but only marginally. Collecting more data was slow and expensive. Active learning helped optimize annotation budget. Transfer learning from a related domain helped when one existed. But the fundamental scarcity remained.

Generative AI offers a new approach. Instead of waiting for real data to arrive, you can generate plausible new examples from the data you have. The technique is not magic; the synthetic examples have limitations covered later in this part. But for many practical problems, generative augmentation increases model accuracy substantially.

The Synthetic Data Pipeline

The standard pipeline has five steps. Start with whatever real labeled data you have (the more the better, but even 50 examples can be useful). Use generative tools (img2img, ControlNet, or text prompts) to produce variations that preserve the labels. Mix the synthetic and real examples in your training data, often with the synthetic outnumbering the real by 5 to 10 times. Train your classifier or detector on the combined dataset. Validate only on real held out examples, never on synthetic.

Choosing the Right Generative Tool

Different generative tools fit different parts of the synthetic data problem. img2img is best when you have existing labeled images and want variations (different lighting, weather, color schemes) that preserve the subject. ControlNet is best when you need to control spatial layout while varying everything else (same pose, different appearance). Pure text to image is best when you need entirely new compositions, though it requires careful prompt engineering to get usable training data.

Style Transfer for Domain Adaptation

A specific case of synthetic data that deserves its own treatment. Domain shift, when your training data and deployment data come from different distributions, is one of the most common causes of CV model failure in production. Your training set is sunny daytime photos. Your deployment sees rain, fog, dusk, and night. The model fails on conditions it never saw.

Generative style transfer through img2img lets you bridge this gap. Take your real training images and apply text prompts that describe target deployment conditions: 'photographed at dusk', 'in heavy fog', 'under harsh fluorescent lighting', 'covered in snow'. The model produces variants of your images that maintain content while changing appearance. Your trained model sees diverse conditions during training and generalizes better to deployment.

INDUSTRY PRACTICE | Autonomous driving was the canary

Autonomous driving teams pioneered generative domain adaptation around 2022. They had millions of miles of sunny daytime training data and were failing in conditions they did not have data for: dawn, dusk, fog, light rain. Synthetic generation of these conditions, applied to existing labeled data, was one of the first large scale industrial uses of generative AI in CV. The same playbook now applies broadly: any team with a domain shift problem should consider generative augmentation before collecting expensive new real data.

Generating Rare Class Examples

Class imbalance is another common CV problem. You have thousands of normal examples and a few dozen examples of rare defects, rare diseases, or rare safety events. Models trained on imbalanced data become biased toward the majority class. Generative models offer a path to balance: generate more examples of the rare class.

The workflow depends on how many real rare examples you have. With even a small number (10 or so), img2img with the rare examples as input plus minor style variations is the safest approach. With more rare examples (50 to 100), ControlNet plus prompts that vary appearance and context can produce diverse variants. With essentially no rare examples but a clear description of what the rare class looks like, text to image generation guided by domain expert prompts can produce candidate examples to seed further data collection.

The Distribution Shift Trap

Now the warnings. Synthetic data is not a free lunch. Generated images look real to humans but have subtle statistical properties that differ from real photographs. These differences are sometimes called distribution shift in the synthetic versus real data domain. A model trained mostly on synthetic data can learn to depend on artifacts of the generation process and fail on real data.

The Validation Rule

The single most important rule in synthetic data work: always validate on real held out data, never on synthetic. If your validation set contains synthetic examples, you have no way to know whether your model has learned to recognize real objects or to recognize generation artifacts. Real validation data is non negotiable.

The Mixing Rule

Mix real and synthetic in training, with real making up as much of the mix as possible. A common pattern is 20 percent real and 80 percent synthetic when real is scarce, but the more real data the better. Models trained on 100 percent synthetic often look great in synthetic tests and fail in real deployment.

> ⚠ **MISCONCEPTION ALERT | Synthetic does not equal real**
>
> The most common mistake teams make with synthetic data is treating it as equivalent to real data. It is not. Synthetic data is for augmentation, not replacement. The validation rule and the mixing rule exist because teams have failed by ignoring them. When a manager says 'why can we not just generate all our training data,' you can explain: because synthetic examples contain subtle artifacts that real photos do not, and a model trained without sufficient real examples will learn to depend on those artifacts.

Quality Control for Synthetic Examples

Not all generated images are usable as training data. Manual review is the gold standard but expensive. Several heuristic checks help. Check that generated examples preserve the correct label (a generated 'defective part' should actually contain a defect). Filter out examples where the generation went wrong (extra fingers, garbled text, impossible geometry). Use diverse prompts to avoid producing many near duplicates that add no information. Consider using a CLIP score (or vision language model score, M08 territory) to filter for prompt fidelity automatically.

KEY TAKEAWAY

Synthetic data is the highest impact workforce use of generative models in CV. The standard pipeline: real examples flow through generation tools (img2img, ControlNet, prompts) to produce synthetic variants, which mix with real in training. Style transfer through img2img enables domain adaptation. Generation of rare class examples balances imbalanced datasets. Two rules are non negotiable: validate only on real data, and mix real with synthetic in training rather than replacing real entirely.

KNOWLEDGE CHECK | Quick check on synthetic data

1. Walk through the five step synthetic data pipeline and name a generation tool that fits each part of it.
2. What is the validation rule for synthetic data work, and why does it matter?
3. Your model is failing on foggy weather conditions. What generative workflow could help, and how?

Part 06 | Limitations, Ethics, and the Legal Landscape

Generative AI tools are powerful. Power without judgment is dangerous. This final part covers what current models still fail at, the ethical concerns every working CV engineer must understand, and the legal landscape as it stands at the time of writing. Treat this part as professional responsibility material, not optional.

What Current Models Still Fail At

Even the best generative models in 2026 have well documented weaknesses. Knowing what the models cannot do is as important as knowing what they can. Promising capabilities that the underlying model cannot deliver is one of the fastest ways to lose credibility.

Counting and Exact Numbers

Generative models are surprisingly bad at counting. Ask for 'five red apples in a bowl' and you may get four or six. Ask for 'exactly three windows' and you get two or four. The model does not have a concept of exact integer count; it produces what looks like a reasonable composition. When precise counts matter, generative models are not the right tool.

Hands and Complex Anatomy

Earlier diffusion models were notorious for producing distorted hands, extra fingers, melted faces, and impossible anatomy. The 2024 and later models (FLUX in particular) have largely fixed this, but older models still struggle. When generating people, always inspect hands and faces carefully. For applications where precise anatomy matters (medical, sports, anything safety relevant), do not rely on generative models for the depiction.

Named Real People

Asking for a named real public figure typically produces a generic person who vaguely resembles a stereotype of that name. Closed APIs (DALL-E 3, Imagen) often refuse such prompts on purpose. Even when open models comply, the result is usually unconvincing. This is both a capability gap and an intentional safety measure.

Highly Specific Technical Content

Schematic diagrams, technical drawings, exact product reproductions, mechanical assemblies, scientific figures. Generative models are not designed for content that requires exact specification. They produce plausible looking content that fails on details. For technical content, generative models are useful for early concept work and useless for final precision.

Deepfakes and Misuse

Convincing synthetic media of real people, often produced without their consent, is now a serious societal problem. Generative models enable cheap, high quality deepfakes for fraud, harassment, political disinformation, and non consensual intimate imagery. The technical capability is widely available; the social and legal infrastructure for handling it is still catching up.

As a CV engineer, you have professional responsibility here. Do not build deepfake systems for unethical purposes. If you work on systems that could be misused (face swapping, voice cloning), apply technical safeguards (watermarking, consent verification, output restrictions). Many CV engineers have refused to work on deepfake adjacent products; that is a legitimate professional choice.

> **⚠ MISCONCEPTION ALERT | Open weights do not absolve responsibility**
>
> Some engineers argue that because open weights models exist, they can build any application on top of them; if they didn't, someone else would. This reasoning is professionally weak. You are responsible for what you build, regardless of the model's accessibility. The fact that someone else might do something harmful is not a license for you to do it. Apply the same ethical standards to generative AI work that you would to any other engineering work.

Watermarking and Provenance

Two major efforts try to make generated images identifiable as such. Invisible watermarking embeds imperceptible markers in generated images that detection tools can extract. Many production generation services apply watermarks automatically. Detection tools are improving but always one step behind generation techniques that can remove or evade watermarks.

Content provenance is the broader effort to attach cryptographic metadata to all media indicating its source and history. The major standard is C2PA (Content Authenticity Initiative), with industry support from Adobe, Microsoft, Sony, BBC, and many others. C2PA metadata travels with the image (when preserved) and lets viewers verify whether an image was generated, where it originated, and what edits were applied.

CAREER CONNECTION | C2PA literacy is becoming required

By 2026, content provenance literacy is moving from specialist topic to professional baseline for CV engineers. Know the name C2PA. Know what it does (cryptographic metadata about image origin and history). Know that camera manufacturers (Sony, Nikon, Leica) are starting to embed C2PA metadata in capture. Know that generation services are also starting to add it. Workflows that strip or preserve C2PA metadata are increasingly part of CV pipeline design.

Dataset Bias and Representational Harm

Generative models reflect their training data. Most large generative models were trained on web scraped image text pairs that overrepresent certain demographics, aesthetics, and cultural perspectives. The result is built in biases. Asked for a 'CEO,' the model may produce primarily white men in suits. Asked for a 'nurse,' it may produce primarily women. Asked for a 'beautiful person,' it may produce a narrow Western beauty stereotype.

These biases cause concrete harms when generative outputs reach real applications. Bias in generated training data can transfer to downstream classifiers. Bias in generated marketing imagery can exclude or stereotype groups. Bias in generated educational content can normalize harmful representations to learners. As a professional CV engineer, test for these biases in your applications and address them when found.

The Legal Landscape

The legal landscape for generative AI is unsettled and rapidly evolving. Two major questions are currently in dispute.

Training Data Copyright

Whether training a generative model on copyrighted images without explicit licensing constitutes fair use or copyright infringement is a contested legal question. Multiple lawsuits are ongoing in the United States, the European Union, and elsewhere. Outcomes vary by jurisdiction. The US legal system is currently leaning toward fair use for transformative use, but this is far from settled. European Union policy under the AI Act is more restrictive.

Output Ownership

Whether AI generated images can be copyrighted, and if so by whom, is also contested. The US Copyright Office has stated that purely AI generated work cannot be copyrighted by the human prompter, but works with significant human creative input can be. Other jurisdictions have different rules. For commercial work, document your creative process to support copyright claims if needed.

Practical Guidance

Until courts and legislatures settle these questions, several practical rules help. For client deliverables, use commercially indemnified services (Adobe Firefly is the clearest example) that guarantee defense and indemnification for copyright claims. For internal use and prototyping, open models are fine, but be cautious about distributing outputs widely. Document your training data sources when using generative output as training data for downstream models. Keep humans in the loop for high stakes outputs (medical, legal, safety relevant).

THINK ABOUT IT | What 'workforce ready' means here

Some engineering topics are pure technique. This one is not. Workforce ready in generative CV in 2026 means technical fluency plus professional judgment. The technical part (load models, write prompts, use img2img and ControlNet) you can learn in this module. The judgment part (when not to use generative AI, how to handle deepfake adjacent requests, what to verify with clients about copyright) is what you build over a career. Take the questions in this part seriously; they will define your professional reputation more than any code you write.

KEY TAKEAWAY

Generative models in 2026 still fail at counting, exact technical content, named real people, and some anatomy details. Deepfake misuse is a serious societal problem; engineers have professional responsibility regardless of model accessibility. Watermarking and C2PA provenance metadata are emerging standards for content authenticity. Dataset bias causes representational harms; test for them. Training data and output copyright are legally unsettled; use indemnified services for high stakes work. Professional judgment matters as much as technical skill.

KNOWLEDGE CHECK | Quick check on responsible use

1. Name three things generative models in 2026 still fail at, and give a workforce implication for each.
2. What is C2PA, and why does its adoption matter for CV engineers?
3. Your client asks for a generated portrait of a named celebrity for a marketing campaign. What considerations would you raise before proceeding?

Review Questions

Work through these on your own first. Then check your answers against the booklet and your notes. If any question feels uncertain, that is the section to reread.

1. Explain in plain English how generative models differ from the analysis models covered in Modules 01 through 06.
2. Name the four workforce use cases for generative models in CV and give an industry example of each.
3. Briefly trace the twelve year arc from GANs in 2014 to current generative tools in 2026, naming three milestones along the way.
4. Compare open weights models and closed API services. Give two strengths of each.
5. Name the two major open weights model families covered in this module and one strength of each.
6. What is Hugging Face Diffusers, and what does the Pipeline abstraction give you?
7. Describe the five principles of prompt engineering. Give one example of how each can change an output.
8. What is a negative prompt, and on which models is it most useful?
9. Describe what each of the three main parameters (sampling steps, guidance scale, seed) controls.
10. Explain the img2img workflow and how the strength parameter affects the output.
11. Walk through how inpainting composes with the segmentation models from Module 06.
12. Name three ControlNet conditioning types and describe what each provides.
13. Describe the synthetic data pipeline for expanding a small labeled dataset. What are the two non negotiable rules?
14. Explain how generative style transfer enables domain adaptation. Give an example.
15. Name three things generative models in 2026 still fail at and give a workforce implication for each.
16. What is C2PA, and why does its adoption matter for CV engineers?
17. Describe two practical rules for handling copyright in client work that uses generative AI.

Glossary

Key terms used throughout this module, in alphabetical order.

C2PA (Content Authenticity Initiative)

Cross industry standard for cryptographic metadata that travels with images to indicate origin, creation method, and edit history. The major effort to provide content provenance for the generative AI era.

ControlNet

A technique for guiding generative models with structural information from a reference image. Common conditioning types include edge maps, depth maps, pose skeletons, and sketches.

DALL-E 3

OpenAI's text to image model released in late 2023. Uses a large language model to interpret prompts. Available through the OpenAI API and ChatGPT.

Deepfake

Convincing synthetic media (image, audio, or video) of a real person, often produced without consent. A growing societal problem enabled by accessible generative AI.

Diffusers (Hugging Face library)

The de facto standard Python library for working with diffusion models. Open source, supports most major open weights models with a unified Pipeline abstraction.

Distribution shift

The phenomenon where training and deployment data have subtly different statistical properties. Synthetic data introduces distribution shift relative to real data; this is why synthetic only validation is invalid.

FLUX

A family of open weights generative models from Black Forest Labs, released in 2024. Notable for fixing common diffusion model weaknesses (hands, text rendering, complex prompts).

GAN (Generative Adversarial Network)

An older generative model architecture that pits a generator against a discriminator. The dominant generative approach from 2014 through the early 2020s, now largely replaced by diffusion models.

Guidance scale (CFG)

Parameter controlling how strictly the model follows the prompt versus how naturally it generates. Higher values stick closer to the prompt; lower values produce more natural images. Typical range 3 to 9.

img2img (image to image)

Generative workflow that transforms an existing image into a new image guided by a text prompt and a strength parameter.

Imagen

Google's text to image model. Powers image generation features across Google products including Search, Workspace, and Vertex AI.

Inpainting

Generative workflow that regenerates a masked region of an image while preserving everything outside the mask. Used for editing, restoration, object removal, and replacement.

LoRA (Low Rank Adaptation)

A lightweight fine tuning technique that adapts a pretrained generative model to a new style or concept with a small number of parameters. Widely used to customize Stable Diffusion.

Midjourney

A closed generative model service known for a distinctive aesthetic. Long ran primarily through Discord; opened web access in more recent versions.

Negative prompt

A second text input describing things to avoid in the generated image. Most useful with Stable Diffusion models; FLUX needs it less; DALL-E 3 does not use it.

Pipeline (Diffusers)

The Diffusers library abstraction that bundles model weights, text encoder, sampler, and postprocessing into a single Python object that handles a specific workflow (text to image, inpainting, etc).

Prompt engineering

The craft of writing text prompts that produce desired outputs from generative models. A primarily experimental skill rather than analytical.

Sampling steps

Parameter controlling how many refinement passes the model takes during generation. More steps means higher quality and longer generation time. Typical range 20 to 50.

Seed

The random starting point for a generation. Same prompt plus same seed produces the same image. Critical for reproducibility.

Stable Diffusion

A family of open weights generative models from Stability AI, released starting in 2022. Major versions include SD 1.5, SDXL, and SD3.

Strength (img2img)

Parameter in img2img controlling how much the model is allowed to deviate from the input image. Low strength preserves the original; high strength acts like text to image.

Super resolution

The task of producing a higher resolution version of a lower resolution input image. Common pretrained models include Real ESRGAN and SUPIR.

Synthetic data

Training data produced by generative models rather than collected from the real world. Used to expand small datasets or generate rare class examples.

Text to image

The basic generative workflow: input text prompt, output generated image. The headline use case for generative AI in vision.

Watermarking

Invisible markers embedded in generated images to enable later identification. Increasingly required by generation services for transparency.

References and Further Reading

The following resources are excellent next steps for deepening your knowledge of generative models for vision. None of them are required for this course.

Foundational Books

Foster, David. Generative Deep Learning, Second Edition. O'Reilly, 2023. The most accessible book length introduction to generative models including GANs, VAEs, diffusion models, and text to image. Worked examples in TensorFlow and PyTorch.

Elgendy, Mohamed. Deep Learning for Vision Systems. Manning Publications, 2020. Covers GANs and visual embeddings as part of broader CV foundations, useful background for generative work.

Goodfellow, Ian; Bengio, Yoshua; and Courville, Aaron. Deep Learning. MIT Press, 2016. Free at deeplearningbook.org. The chapter on generative models covers the older theoretical foundations.

Recent Books (2024 to 2026)

Bansal, Atul Garg and Gupta, Prashant. Mastering Generative AI for Computer Vision. Springer, 2024. Comprehensive coverage of GANs, VAEs, diffusion models, and applications including style transfer, super resolution, and synthetic data.

Ayyadevara, V Kishore and Reddy, Yeshwanth. Modern Computer Vision with PyTorch, Second Edition. Packt Publishing, June 2024 (ISBN 9781803231334). Includes detailed chapters on diffusion models, GANs, and generative applications in CV.

Chollet, Francois. Deep Learning with Python, Third Edition. Manning Publications, 2025. Covers generative deep learning from a Keras and applied perspective.

Foundational Papers

Goodfellow, Ian et al. Generative Adversarial Nets. NeurIPS, 2014. The original GAN paper.

Ho, Jonathan; Jain, Ajay; and Abbeel, Pieter. Denoising Diffusion Probabilistic Models. NeurIPS, 2020. The DDPM paper that established the modern diffusion model framework.

Rombach, Robin et al. High Resolution Image Synthesis with Latent Diffusion Models. CVPR, 2022. The Latent Diffusion paper that became Stable Diffusion.

Saharia, Chitwan et al. Photorealistic Text to Image Diffusion Models with Deep Language Understanding. NeurIPS, 2022. The Imagen paper from Google.

Zhang, Lvmin; Rao, Anyi; and Agrawala, Maneesh. Adding Conditional Control to Text to Image Diffusion Models. ICCV, 2023. The ControlNet paper.

Esser, Patrick et al. Scaling Rectified Flow Transformers for High Resolution Image Synthesis. ICML, 2024. The SD3 paper introducing rectified flow architectures.

Documentation and Tutorials

Hugging Face Diffusers documentation at huggingface.co/docs/diffusers. The primary reference for working with open generative models in Python. Includes tutorials for every major workflow.

Stability AI documentation at platform.stability.ai/docs. Covers all Stable Diffusion versions and their API.

Black Forest Labs FLUX documentation. Covers the FLUX model family and recommended usage patterns.

OpenAI DALL-E 3 API documentation at platform.openai.com/docs. Covers prompt usage and API integration.

Content Authenticity Initiative at contentauthenticity.org. The home for the C2PA standard and adoption news.

Adobe Firefly documentation at helpx.adobe.com/firefly. Covers Firefly's commercial safety guarantees and integration.

Looking Ahead

Module 08 brings you to the architectural revolution that has reshaped computer vision starting around 2020: transformers. You will learn what Vision Transformers (ViT) are, how they differ from the CNNs that dominated Modules 01 through 06, and why they took over many CV tasks within a few years of their introduction.

More importantly for the workforce of 2026, M08 covers Vision Language Models (VLMs): systems that take both images and text as input and produce text as output. GPT-5 vision, Claude Opus 4.7, Gemini 3, Pixtral, Qwen2.5-VL, and others. These models can describe images, answer questions about them, and ground actions in what they see. They are the foundation for almost every modern AI agent that interacts with the visual world.

Module 09 then builds on M08 to cover AI agents for computer vision, where VLMs become the eyes of broader autonomous systems. Module 10 covers video analysis and object tracking. Module 11 wraps up with model optimization and deployment, plus a course wide review. The remaining modules will build steadily on everything you have learned so far. Half the semester behind you, half ahead. Keep going.

email or message me if you have any questions